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Sang Hyeok Im (임상혁)

Curriculum Vitae

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea — sanghyeok.im97@gmail.com

EDUCATION

Ph.D student in Astronomy Program (2021.02 - present)

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea

B.S. in Physics (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

B.S. in Astronomy and Atmospheric Sciences (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

RESEARCH INTERESTS

1. Large-scale structures of the universe
2. Observational cosmology
3. Environmental effects on galaxy properties
4. Formation and evolution of proto-clusters

PUBLICATIONS

Refereed Journal Article

First Author

1. Testing Ly α Emitters and Lyman-Break Galaxies as Tracers of Large-Scale Structures at High Redshifts
Sang Hyeok Im et al.
2024, ApJ, 972, 196 (NASA ADS)
Highlighted as AAS Nova¹ (see here)
2. ODIN: Rest-frame Optical Morphologies and Star Formation Activity of Ly α Emitters at $z = 2.4, 3.1$, and 4.5
Sang Hyeok Im et al.,
2026, accepted for publication in ApJ (arXiv:2605.29344)

Second Author

1. Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies
Gene Yun, **Sang Hyeok Im** et al.,
2026, ApJ, 1007, 168 (arXiv:2607.17271)

Co-Author

1. ODIN: Searching for LyC emission from Lyman- α emitters at $z = 4.5$ in the E-COSMOS and XMM-LSS fields
Eunsuk Seo et al., (**Sang Hyeok Im**)
2026, ApJ, 1001, 224 (NASA ADS)
2. ODIN: Clustering Properties of Ly α Blobs at $z \sim 2.4$ and 3.1
Byeongha Moon et al., (**Sang Hyeok Im**)

¹AAS Nova is the research highlights from the journals of the American Astronomical Society.

2026, ApJ, 1004, 122 (NASA ADS)

3. ODIN: A New Lyman Alpha Blob Selection Method, Sample, and Statistical Analysis at $z \sim 3.1$
Byeongha Moon et al., (**Sang Hyeok Im**)
2025, ApJ, 998, 345 (NASA ADS)
4. ODIN: Characterizing the Three-dimensional Structure of Two Protocluster Complexes at $z=3.1$
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2025, accepted for publication in ApJ (arXiv:2511.11826)
5. ODIN: Star Formation Histories Reveal Formative Starbursts Experienced by Ly α -emitting Galaxies at Cosmic Noon
Nicole M. Firestone et al., (**Sang Hyeok Im**)
2025, ApJL, 986, 8 (NASA ADS)
6. ODIN: High Clustering Strength of Protoclusters at Cosmic Noon
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2025, ApJ, 982, 74 (NASA ADS)
7. ODIN: Identifying Protoclusters and Cosmic Filaments Traced by Ly α -emitting Galaxies
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)
2024, ApJ, 977 119 (NASA ADS)
8. ODIN: Improved Narrowband Ly α Emitter Selection Techniques for $z = 2.4, 3.1$, and 4.5
Nicole M. Firestone et al., (**Sang Hyeok Im**)
2024, ApJ, 974, 217 (NASA ADS)
9. The One-hundred-deg² DECam Imaging in Narrowbands (ODIN): Survey Design and Science Goals
Kyoung-Soo Lee et al., (**Sang Hyeok Im**)
2024, ApJ, 962, 36 (NASA ADS)
10. ODIN: Where Do Ly α Blobs Live? Contextualizing Blob Environments within the Large-Scale Structure
Vandana Ramakrishnan, Byeongha Moon, **Sang Hyeok Im** et al.,
2023, ApJ, 951, 119 (NASA ADS)

GRANTS AND FELLOWSHIPS

CIAS Student Fellow (2021.09 - 2026.02)

- o Institutional Location: Korea Institute for Advanced Study, 85 Hoegi-ro, Dongdaemun-gu, Seoul 02455, Republic of Korea
- o Project Title: A 100 Deg² DECam Narrow-band Survey for the LSST Era: Tracing the Largest Cosmic Structures in the Distant Universe

PRESENTATIONS

Oral Presentations

1. **110th KAS Fall Meeting** (2024 Oct. 16 - 18 @ Gangneung, Korea)
Title: Testing LAEs and LBGs as Tracers of Large-Scale Structures at High Redshifts
2. **K-GMT Science Program Users Meeting 2024** (2024 Aug. 21 - 23 @ KASI, Daejeon, Korea)
Title: Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies
3. **9th Galaxy Evolution Workshop** (2023 Feb. 20 - 23 @ Kyoto University, Japan)
Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in the high-redshift universe

Poster Presentations

1. **ELT Science in Light of JWST** (2024 June 3 - 7 @ Tohoku University, Sendai, Japan)
Title: Large-scale structures in the universe traced by Lyman alpha emitters and Lyman-break galaxies
2. **108th KAS Fall Meeting** (2023 Oct. 18 - 20 @ Jeju, Republic of Korea)
Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of the large-scale structures in the high-z universe
3. **XXXIth General Assembly International Astronomical Union** (2022 Aug. 2 - 11 @ Busan, Korea)
Title: Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in a simulated universe

RESEARCH EXPERIENCES

Observing Proposal (PI)

1. **Gemini/N GMOS-longslit (2023B)**
Title: Studying the Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies
Allocated Time: 10.1 hr
Results are published in ApJ (as a second author; see the publication list)

Observing Proposal (CoI)

1. **MMT/Hectospec (2024A - 2026A)**
Title: Revisiting the Faint-end Cutoff in the Galaxy Luminosity Function of the Satellites around Isolated Early-type Galaxies
Allocated Time: total 7 nights

REFERENCES

Prof. Ho Seong Hwang
Seoul National University, Korea
galaxy79@snu.ac.kr

Prof. Changbom Park
Korea Institute for Advanced Study, Korea
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